PROJECT REPORT

SOFTWARE INSTALLATION REQUEST

ServiceNow Service Portal | Flow Designer Automation

Project Details

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Project Platform	ServiceNow Service Portal
Academic Year	2026–2027
ServiceNow release / scope	Australia release / Global application scope
Catalog Item	Software Installation Request
Automation	Flow Designer; Legacy Workflow is not used

Project Summary

The project provides employees with a structured way to request company-approved software. The catalog form captures software name, version, business reason, and urgency. Flow Designer routes the Requested Item to the requester's manager, creates a fulfillment Catalog Task for Software Support Team after approval, monitors task status, and sends an outcome email. A separate Business Rule creates an Incident when a Requested Item is placed On Hold for a license issue.

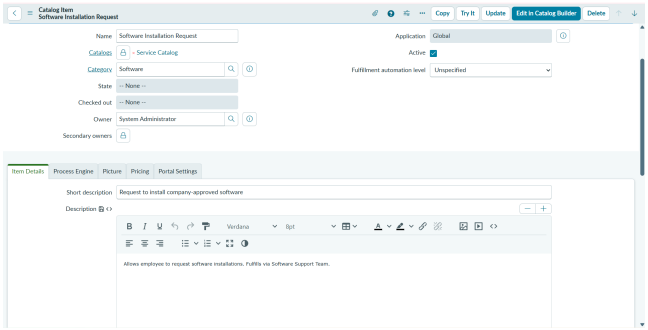


Figure 1. Software Installation Request catalog item configuration.

1. INTRODUCTION & IDEATION

1.1 Project Overview

Employees may need approved software to complete daily work. A request requires the correct software and version, a business justification, manager approval, support-team fulfillment, and confirmation of the result. License availability can delay installation. This project connects these steps through one ServiceNow Service Catalog item and a Flow Designer process.

1.2 Purpose

The purpose is to standardize requests, collect complete details at submission, require manager approval before fulfillment, assign approved work to Software Support Team, monitor task outcomes, and communicate either successful installation or a license-related delay. The separate Incident automation records license issues on the Requested Item.

1.3 Customer Problem Statement

I am: An employee who needs software installed on a work computer to perform daily tasks or job responsibilities.
I'm trying to: Submit a request with software name, version, business reason, and urgency, then receive the software after approval.
But: I may not know the correct process, may provide incomplete information, may wait for manager approval, or may not know whether a license is available.
Because: Requests require business approval and support-team fulfillment; license availability affects installation timing.
Without a structured process, details, ownership, and progress are difficult to follow.
Which makes me feel: Uncertain, delayed in completing work, and inconvenienced by follow-ups.

1.4 Stakeholder Problem Statements

PS-1	Employee needing work software	Needs to request software and receive the correct version, but may not know request status, approval progress, or license availability.	Uncertain and delayed at work.
PS-2	Manager or Software Support Team member	Needs to review, approve, assign fulfillment, and communicate results; incomplete details and follow-ups make ownership harder to track.	Rework, unclear responsibility, and employee delays.

1.5 Empathy Map Summary

Think & Feel	Wants the right software; worries about approval and license delays.	Clear request path and updates.
Hear	May hear requirements from a manager or questions from support.	Collect details at submission.
See	Sees catalog form and email updates; approval/fulfillment happen behind the scenes.	Structured form and outcome emails.
Say & Do	Submits request, enters details, waits for approval and fulfillment.	Route work in the correct order.
Pain / Gain	Pain: unclear progress, repeated follow-ups, delays. Gain: one channel, clear ownership, confirmation.	Track outcomes and explain delays.

The empathy map is a project-design hypothesis based on identified needs, not a claim of formal employee interviews or survey research.

2. BRAINSTORMING & REQUIREMENTS

2.1 Brainstorming and Prioritization

Ideas were grouped around the implemented request-to-fulfillment process. Request intake, approval, task creation, task monitoring, and requester notifications were prioritized as High. Incident automation and Update Set packaging were treated as Medium supporting capabilities.

1	Catalog item captures software, version, business reason, and urgency.	Request Intake	High
2	Route Requested Item to requester’s manager for approval.	Approval	High
3	Create Catalog Task and assign it to Software Support Team.	Fulfillment	High
4	Wait for Closed Complete or Pending task state.	Task Monitoring	High
5	Send personalized completion or license-unavailable email.	Notifications	High
6	Create Incident from Requested Item when On Hold for license issue.	Exception Handling	Medium
7	Capture configuration in Update Set and test request paths.	Deployment & Testing	Medium

2.2 Catalog Item Configuration

The active catalog item is Software Installation Request, in the Software category and Service Catalog. Its short description is “Request to install company-approved software.” Its description states that employees can request software installations and fulfillment is via Software Support Team.

What software do you need?	Single Line Text	Captures application name. Internal name shown: Software_Name.
Specify version (if required).	Single Line Text	Captures requested version.
Why do you need this software?	Multi Line Text	Captures business reason.
Select urgency level.	Select Box	Captures requester’s urgency selection.

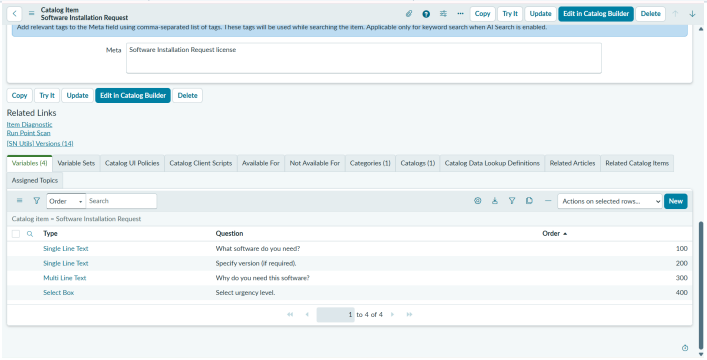


Figure 2. Catalog item’s four variables and their configured order.

3. PROJECT DESIGN & FLOW ARCHITECTURE

3.1 Proposed Architecture

The Service Catalog is the employee entry point. Requested Item [sc_req_item] stores the request. Flow Designer coordinates manager approval and fulfillment. Catalog Task [sc_task] represents support work. Email actions notify the requester. A separate Business Rule creates an Incident for the Requested Item On Hold license-issue condition. The Update Set packages configuration changes.

Service Catalog	Captures software name, version, business reason, and urgency.
Requested Item [sc_req_item]	Stores request and requester context.
Flow Designer	Runs approval, task creation, variable retrieval, wait, branching, and notifications.
Catalog Task [sc_task]	Tracks support fulfillment assigned to Software Support Team.
Business Rule / Incident	Creates an Incident when Requested Item is On Hold for license issue.
Update Set	Captures configuration for transfer between instances.

3.2 Software Request Workflow

The flow is named Software Request Workflow, runs in the Global application scope, and starts with the Service Catalog / Catalog Item Requested trigger. It uses Flow Designer; Legacy Workflow is not used.

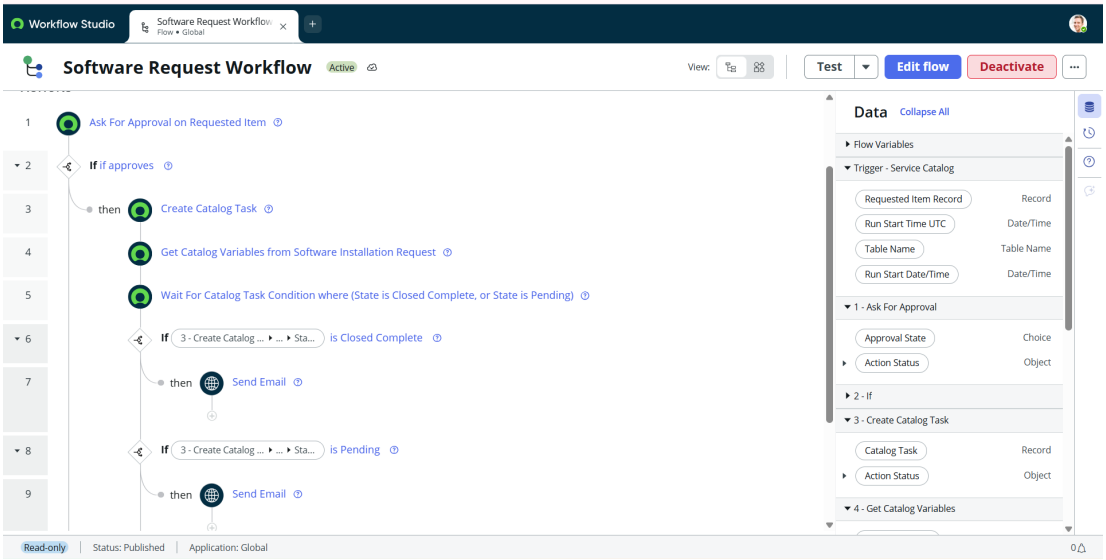


Figure 3. Latest workflow layout: Ask For Approval, approved branch, Create Catalog Task, Get Catalog Variables, Wait For Catalog Task Condition, and two Send Email branches.

3.3 Flow sequence

1	Catalog Item Requested trigger	Starts when the catalog item is submitted.
2	Ask For Approval on Requested Item	Sends approval to requester's manager.
3	If approval is Approved	Only approved requests continue.
4	Create Catalog Task + Get Catalog Variables	Creates task and exposes catalog variables to later actions.
5	Wait For Catalog Task Condition	Waits for Closed Complete OR Pending.
6	If / Send Email branches	Notifies requester of the corresponding outcome.

4. FLOW CONFIGURATION & EMAILS

4.1 Manager Approval

Ask For Approval uses Trigger → Requested Item Record, table Requested Item [sc_req_item], Approval as the approval field, and Approval history as the journal field. The rule is Anyone approves; the approver is the requester's manager. The flow proceeds only when Approval State is Approved.

4.2 Catalog Task Creation

Create Catalog Task writes to Catalog Task [sc_task], links the task to the Trigger → Requested Item Record, sets Short Description to Software Fulfillment Task, and assigns it to Software Support Team. The action's Wait option is OFF; a separate Wait For Condition monitors the task.

4.3 Wait and Branch Conditions

Wait For Catalog Task Condition references the Catalog Task output from Create Catalog Task and uses Catalog Task [sc_task]. It waits for State = Closed Complete OR Pending. The next If branches check the same task state: Closed Complete routes to the success email; Pending routes to the license-unavailable email.

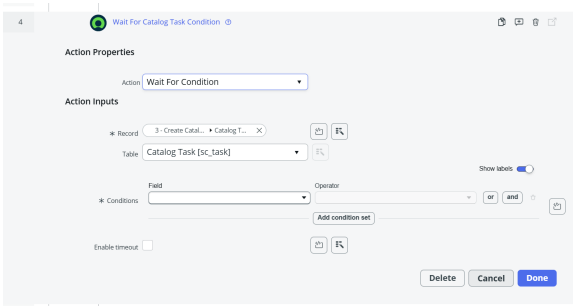


Figure 4. Wait For Condition uses the created Catalog Task record and Catalog Task [sc_task].

Project-specific convention: Catalog Task state choices did not include On Hold. Pending is used only in this project as the license-unavailable signal; it is not a universal meaning of Pending. The separate Business Rule uses Requested Item state On Hold.

4.4 Personalized Email Design

Both Send Email actions target Requested Item [sc_req_item] and use Requested for → Email as the recipient. The requester name is available from Requested for → Name. Get Catalog Variables was added to make the four catalog variable values available to subsequent actions.

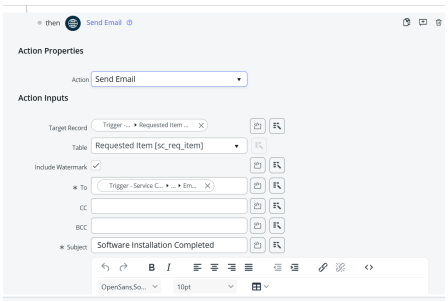


Figure 5. Send Email configuration showing Requested Item target, requester email pill, and completion subject.

Success subject: Software Installation Completed. Intended message confirms the requested software license has been assigned and installed, with requester name and software details.

Shortage subject: Software License Unavailable. Intended message explains no license is currently available, a new license must be obtained, and expected installation time is approximately one week.

Verification note: Get Catalog Variables is visible in the latest workflow screenshot, but final variable-to-email-body mapping and successful delivery of both personalized email branches were not demonstrated. They require final end-to-end verification.

5. LICENSE-ISSUE INCIDENT & UPDATE SET

5.1 Business Rule Configuration

The Business Rule is named Create Incident on License Issue. It is in Global scope, applies to Requested Item [sc_req_item], and has Active and Advanced checked. It runs After; Insert and Update are selected. The condition is the Requested Item entering On Hold for a license issue (state value 3 in the script).

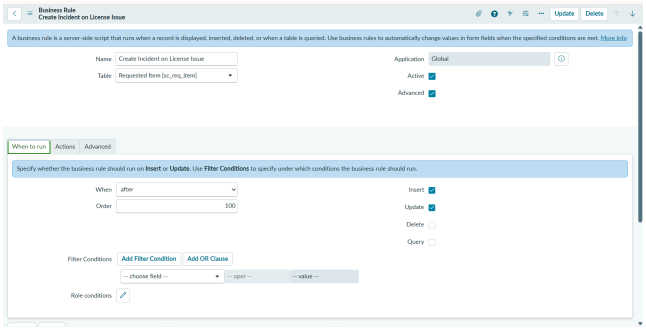


Figure 6. Business Rule record showing table, Active/Advanced, After, Insert and Update settings.

5.2 Business Rule Script

```
if (current.state == 3) { // On Hold
var inc = new GlideRecord('incident');
inc.initialize();
inc.short_description = 'License unavailable for ' + current.request_item;
inc.description = 'Software request could not be fulfilled due to license shortage.';
inc.caller_id = current.request.requested_for;
inc.assignment_group.setDisplayValue('Software Support');
inc.priority = 3;
inc.insert();
}
```

The script checks state 3, initializes an Incident, sets its short description and description, derives caller from the Requested Item's request/requested-for reference, assigns the group by display value, sets priority 3, and inserts the record. The script uses assignment group display value Software Support; the Catalog Task flow uses Software Support Team. These labels are preserved as shown and should be reconciled against the intended group record.

5.3 Update Set

The Update Set is named Software Installation Request Setup, scope Global, state In progress, and its record shows 12 Customer Updates. Captured updates include catalog variables and other configuration records. The project discussion records that the project was also verified in a second instance (Instance B); the screenshot itself shows the Update Set as In progress, not Committed.

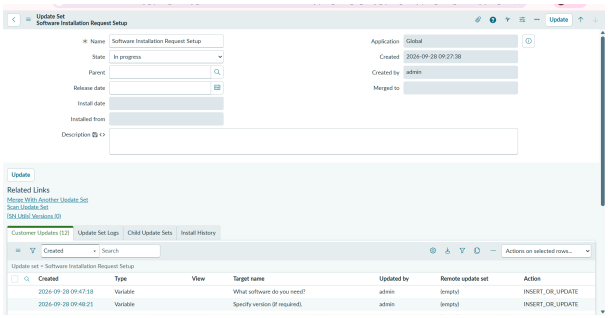


Figure 7. Update Set details and Customer Updates count.

6. FUNCTIONAL TESTING & OBSERVED RESULTS

6.1 Submitted Requested Item

The visible test record is Requested Item RITM0010003, linked to Request REQ0010003. Item: Software Installation Request. Requested for: System Administrator. Stage: Request Approved. State: Open.

Software	Microsoft Office 365
Version	2021
Business reason	Required for Daily work
Urgency	High
RITM / REQ	RITM0010003 / REQ0010003

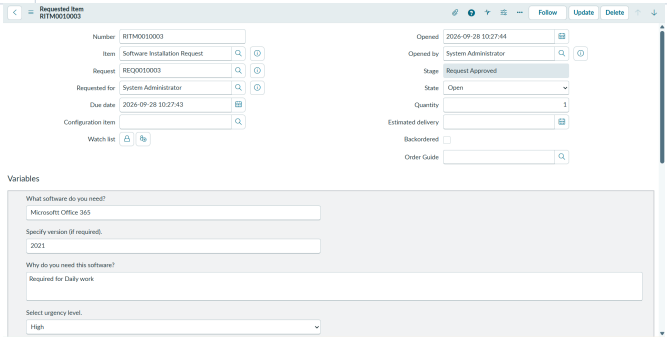


Figure 8. Requested Item RITM0010003 and submitted catalog variable values.

6.2 Approval and Fulfillment Task

The approval related list for RITM0010003 shows an Approved record for Abraham Lincoln and a separate No Longer Required record. The fulfillment task is SCTASK0010007, linked to RITM0010003, with Short Description Software Fulfillment Task, Assignment group Software Support Team, Requested for System Administrator, and State Open.

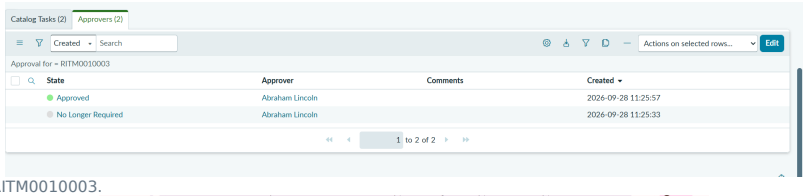


Figure 9. Approval records for RITM0010003.

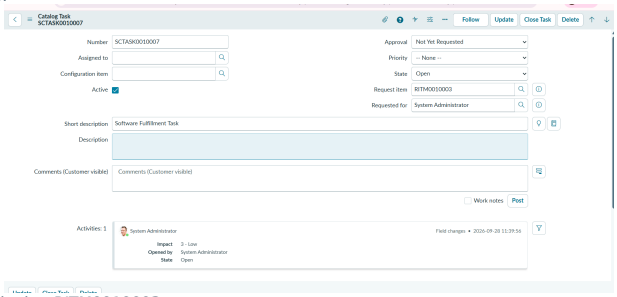


Figure 10. Catalog Task SCTASK0010007 linked to RITM0010003.

7. TEST MATRIX, CONCLUSION & FUTURE SCOPE

7.1 Flow Execution Evidence

A Flow Designer execution screenshot shows Test Run – Completed for an earlier version of Software Request Workflow. The Service Catalog trigger, Ask For Approval action, Approved If branch, and Create Catalog Task action completed. This supports the original approval-to-task path. It predates the later Get Catalog Variables, Wait For Condition, and email-branch enhancements; it does not prove the final email branches were tested.

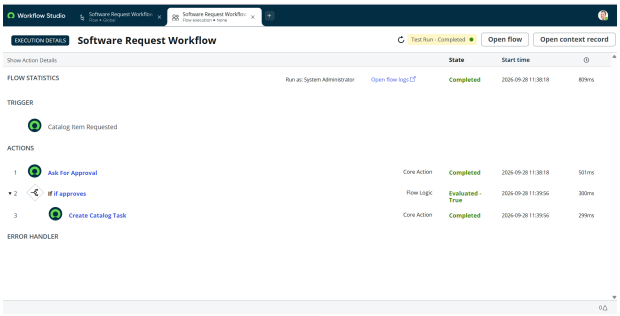


Figure 11. Completed earlier flow execution showing approval and task creation.

7.2 Functional Test Matrix

Approved request	Submit item and manager approves; flow creates Catalog Task for Software Support Team.	Core path evidenced by RITM, approval, task, and earlier completed run.
Successful installation	After Wait For Condition is active, set task to Closed Complete; success email should be sent.	Test procedure defined; final branch and email delivery not evidenced.
License unavailable	After wait is active, set task to Pending; email should explain new license and ~one-week estimate.	Test procedure defined; final branch and email delivery not evidenced.
Incident automation	Move Requested Item to On Hold; Business Rule should insert Incident.	Rule and script recorded; generated Incident evidence not supplied.

Testing note: submit a real Service Catalog request, approve it, and wait until the flow reaches Wait For Condition before changing the created Catalog Task to Closed Complete or Pending. Changing the task before the flow is waiting may bypass the intended observation of the state transition.

7.3 Results

Evidence confirms the catalog item and four variables; a submitted RITM with Microsoft Office 365 / 2021 / Required for Daily work / High; manager approval; creation of a task assigned to Software Support Team; a completed earlier approval-to-task flow run; the later flow layout with Get Catalog Variables, Wait For Condition, and two email branches; Business Rule configuration; and an Update Set record.

7.4 Advantages, Limitations & Conclusion

Advantages: one request channel, complete intake details, approval before fulfillment, clear task ownership, outcome monitoring, requester email design, license-issue Incident automation, and Update Set packaging.
Limitations: Pending is a project-specific shortage signal; the Business Rule group display value “Software Support” differs from the flow’s “Software Support Team”; final personalized email mapping/delivery and generated Incident evidence remain to be verified.
Conclusion: The project establishes a structured ServiceNow request, approval, and fulfillment process. The available evidence supports the catalog configuration, submitted request, approval, task creation, earlier workflow execution, Business Rule setup, and Update Set. Final email branches and Incident creation should be end-to-end tested before claiming complete functional validation.

7.5 Future Scope

Use a dedicated license-availability field or explicit fulfillment outcome instead of overloading Pending; standardize the Software Support group reference; add timeout/escalation handling; track license procurement and delivery dates; and capture final execution logs for both email branches and the generated Incident.

Appendix: the latest workflow screenshot includes Get Catalog Variables and both outcome branches. The variable definition record below shows “What software do you need?” and internal name Software_Name.

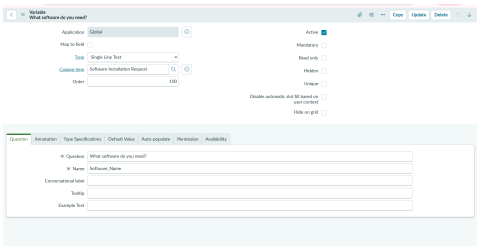


Figure 12. Catalog variable definition record for Software_Name. Report basis: project-details table, brainstorming and prioritization, problem statement, empathy map, ServiceNow screenshots, and configuration details shared in the conversation. No student ID or team ID has been invented.